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Studierichting: Informatica
Oefeningenreeks 4A nr. 38.5

Bereken de volgende onbepaalde integraal $\int \frac{x+4}{x^2 - x + 1dx}$

$$= \int \frac{\frac{1}{2}(2x-1) + \frac{9}{2}dx}{x^2 - x + 1}$$

$$= \frac{1}{2}\int \frac{2x - 1dx}{x^2 - x + 1} + \frac{9}{2}\int \frac{dx}{x^2 - x + 1}$$

$u = x^2 - x + 1, du = 2x - 1dx, v = x - \frac{1}{2}, dv = dx$

$$= \frac{1}{2}\int \frac{du}{u} + \frac{9}{2}\int \frac{dv}{v^2 + \frac{3}{4}}$$

$t = \frac{2v}{\sqrt{3}}, dt = \frac{2dv}{\sqrt{3}}$

$$= \frac{1}{2}\ln|u| + \frac{9}{2} \cdot \frac{\sqrt{3}}{2}\int \frac{dt}{\left(\frac{\sqrt{3}t}{2}\right)^2 + \frac{3}{4}}$$

$$= \frac{1}{2}\ln|u| + \frac{9\sqrt{3}}{4}\int \frac{dt}{\frac{3}{4}(t^2+1)}$$

$$= \frac{1}{2}\ln|u| + \frac{9\sqrt{3}}{4} \cdot \frac{4}{3}\int \frac{dt}{t^2+1}$$

$$= \frac{1}{2}\ln|u| + 3\sqrt{3}\,\mathrm{Bgtan}\, t + C$$

$$= \frac{1}{2}\ln|x^2 - x + 1| + 3\sqrt{3}\,\mathrm{Bgtan}\left(\frac{2x-1}{\sqrt{3}}\right) + C$$

References